MODULE 3 · MACHINE LEARNING · DAY 34

More Than One Column

Multiple linear regression — and the trap that comes with it.

5	16	1
sections	worked pictures	number per column



Boston University

Prepared & presented by **Srinivasa Sai Chava**

WHERE WE ARE

Yesterday one column. Today, as many as you like.

1 Day 33

One input column. Find the slope and the intercept.

2 TODAY

Many input columns. Find one number for each of them.

3 The catch

The numbers become much harder to READ correctly.

Everything from yesterday still holds

The error is still actual − prediction. We still square the errors and add them up. We still find the numbers that make that total smallest. R^2 still compares the model with guessing the average. And you must still split before you fit, still check the errors for a pattern, and still refuse to predict outside your data.

The maths is a small step. The hard part today is INTERPRETATION — what one coefficient means when five other columns are in the room with it.

SECTION 1 OF 5

Adding a second column

One number per column, and nothing else changes.

- Six flats, two facts
- A line becomes a plane
- Each column alone
- Who finds the numbers?

THE PROBLEM

Six flats. This time you know two things about each.

EXAMPLE

Yesterday you had area alone. Now you also know how many bedrooms each flat has.

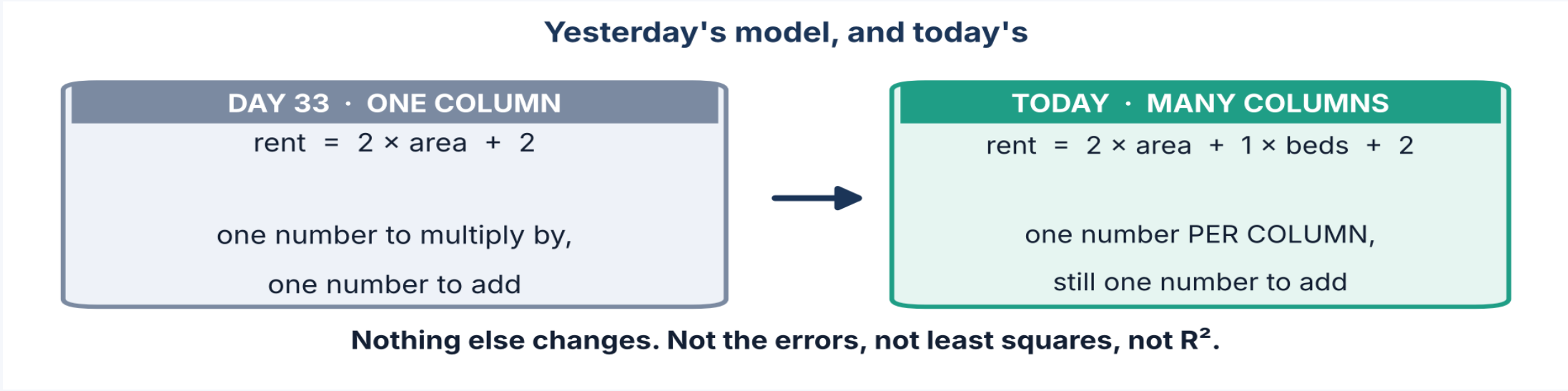
flat	area (100s of sqft)	bedrooms	rent (₹ thousands)
1	5	1	14
2	7	1	16
3	9	2	21
4	11	2	27
5	13	3	31
6	9	3	23

Look at flats 3 and 6: the same 900 sqft, but one has two bedrooms and the other three — and the rents differ by ₹2,000.

Area alone cannot tell those two flats apart. If bedrooms matter at all, we need a model that can use both.

THE IDEA

One number per column. That is the whole change.

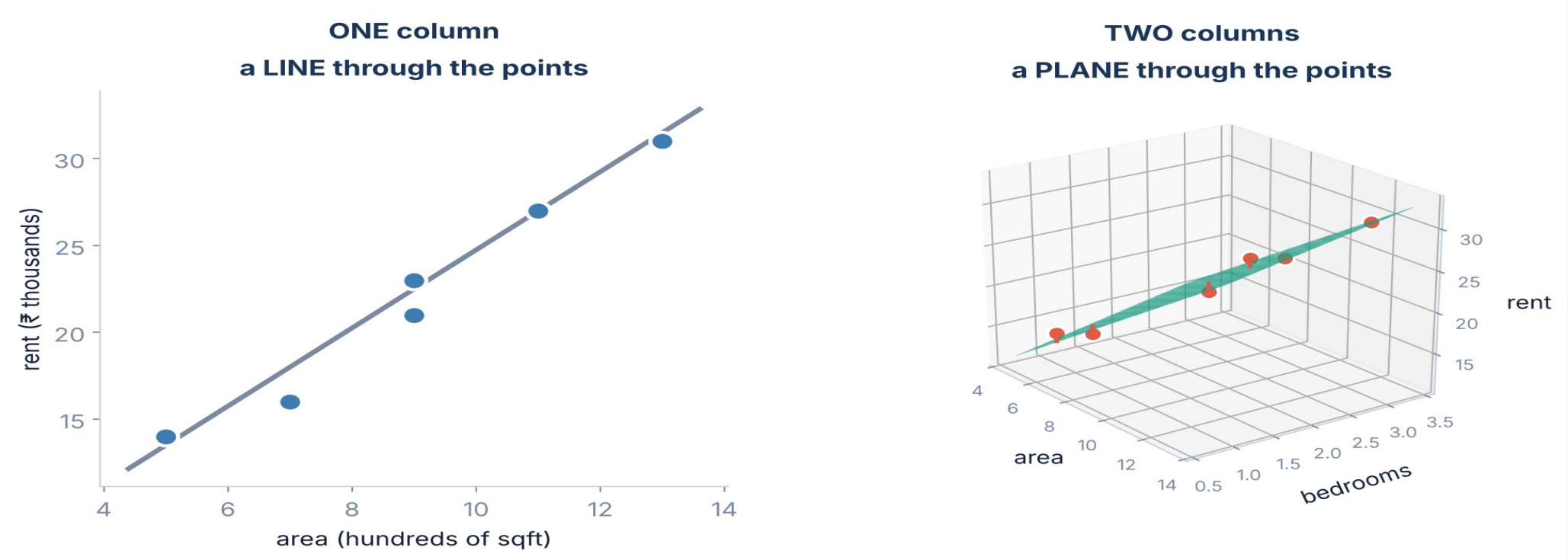


In general: $\text{answer} = w_1 \times \text{column}_1 + w_2 \times \text{column}_2 + \ldots + w_n \times \text{column}_n + \text{intercept}$

Ten columns means ten of those w numbers, plus one intercept. Eleven numbers, and the model is complete.

WHAT IT LOOKS LIKE

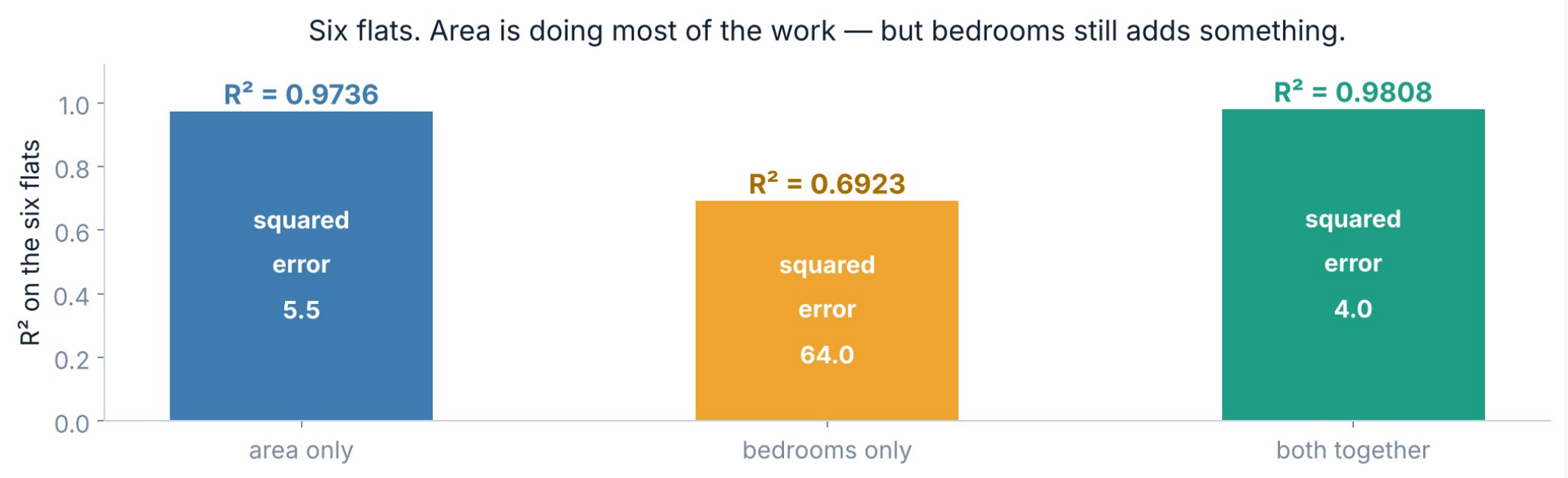
A line becomes a flat sheet



With three columns it becomes a shape in 4 dimensions that nobody can draw. The arithmetic does not care — but this is the last slide where you can SEE the model.

DOES THE SECOND ONE HELP?

Each column alone, and the two together



Area alone already gets 0.9736. Adding bedrooms takes it to 0.9808 — a real improvement, but a small one. Most of the signal was in area all along.

BY HAND

The best numbers, and how to check them

$$\text{rent} = 2 \times \text{area} + 1 \times \text{bedrooms} + 2$$

area	beds	2 × area	1 × beds	+ 2	model says	actual	error
5	1	10	1	2	13	14	+1
7	1	14	1	2	17	16	−1
9	2	18	2	2	22	21	−1
11	2	22	2	2	26	27	+1
13	3	26	3	2	31	31	0
9	3	18	3	2	23	23	0

errors add to 0

as always, for the best fit

→

squares add to 4

nothing beats this

→

$R^2 = 0.981$

that is 1 − 4/208

FINDING THE NUMBERS

So who works out the 2, the 1 and the 2?

THE FORMULA

Still exists — but it is now a matrix calculation, not something you do on paper. sklearn runs it for you, and it is still exact.

GRADIENT DESCENT

Works unchanged. Yesterday you nudged two numbers downhill. Now you nudge as many as you have columns — the loop is identical.

```
X = flats[["area", "bedrooms"]]        # a TABLE with two columns now
model = LinearRegression().fit(X, rent)
print(model.coef_, model.intercept_)   #  [2. 1.]   2.0
```

The only thing that changed in the code is that X has two columns instead of one. The .coef_ attribute now returns a list — one number per column, in the order you supplied them.

SECTION 2 OF 5

What each number means

Where the real difficulty starts.

- Holding things constant
- Reading a real model
- One flat, by hand
- When a sign flips

READING THE MODEL

A coefficient is a promise about two flats

What "+₹23 per square foot" actually promises

FLAT A	1,000 sqft · 2 beds · 8 yrs · 2 km
FLAT B	1,001 sqft · 2 beds · 8 yrs · 2 km

rent differs by ₹23

identical in every column except one

The coefficient compares two flats that differ in ONE column and match on every other.
That is why it changes when you add or remove a column — you have changed what is being held still.

"+₹23 per square foot" does NOT mean bigger flats cost ₹23 more per foot. It means: take two flats identical in bedrooms, age and distance, and the extra foot is worth ₹23.

READING THE MODEL

A real model: 500 flats, four columns

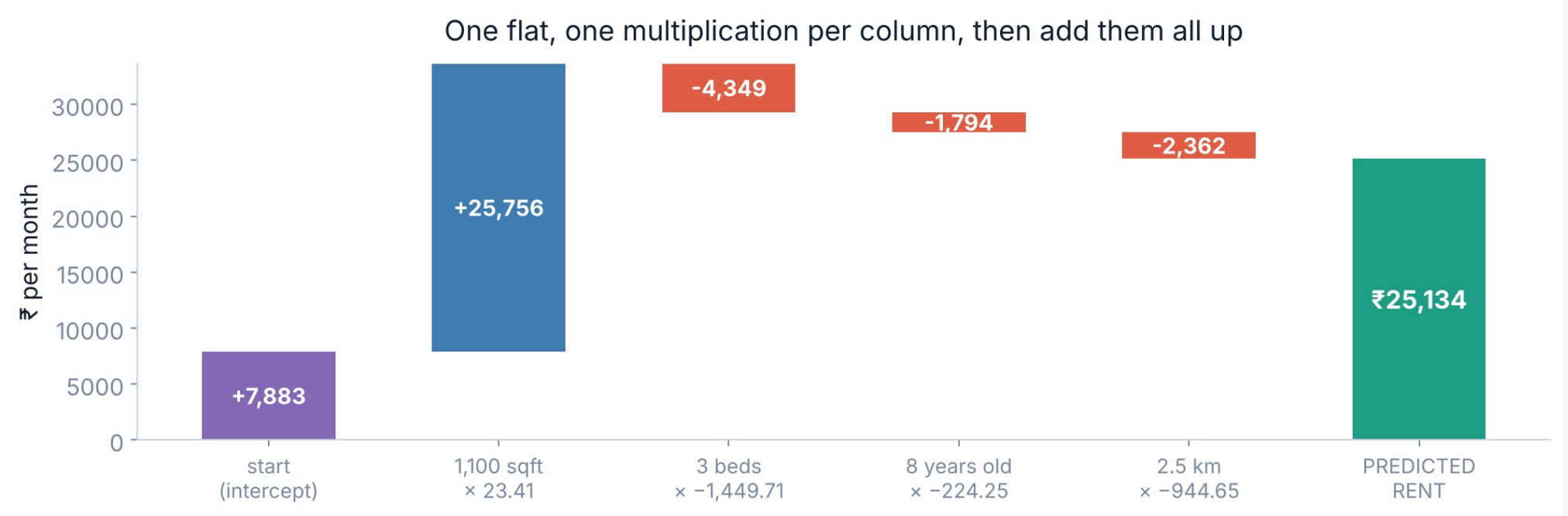
column	coefficient	what it says, in English
`sqft`	+23.41	one more square foot, everything else equal, is worth ₹23 a month
`bedrooms`	−1,449.71	one more bedroom AT THE SAME AREA costs ₹1,450 a month
`age_years`	−224.25	each year of age knocks ₹224 off the rent
`metro_km`	−944.65	each kilometre further from the metro costs ₹945
`intercept`	+7,882.88	where the arithmetic starts — not a flat anybody could rent

R^2 on rows it studied 0.835 · on rows it never saw 0.811 · average miss ₹2,288 a month

Read the bedrooms row again. It is NEGATIVE — more bedrooms, less rent. That cannot be right, and the next four slides are about why it is.

USING THE MODEL

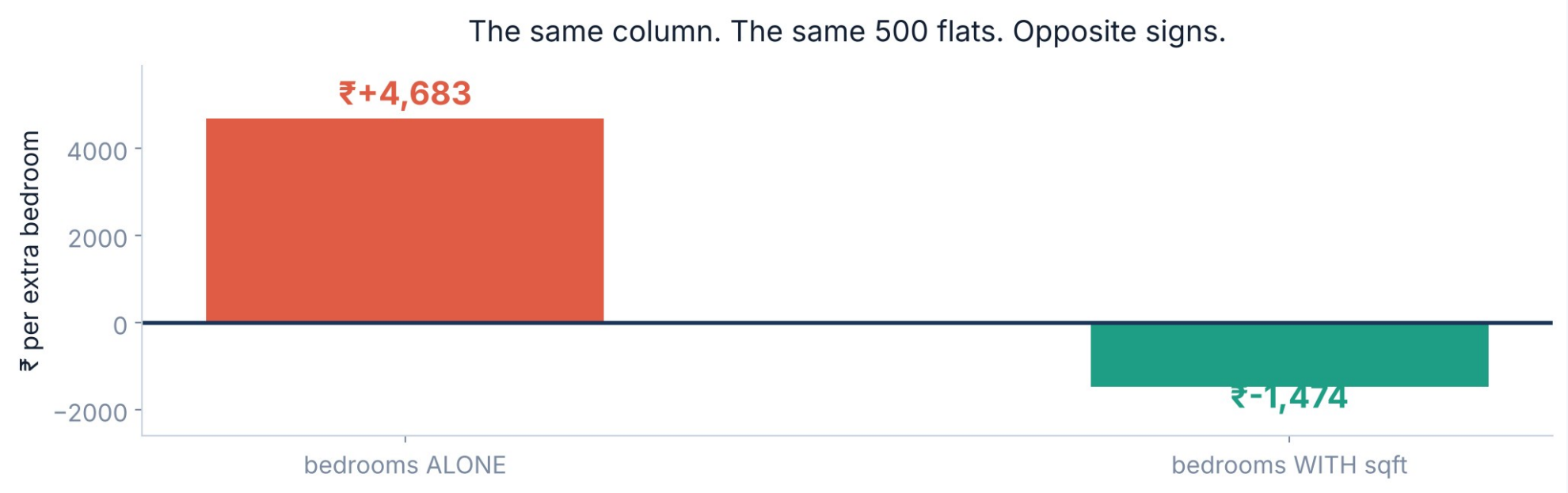
Predicting one flat: multiply, multiply, add



1,100 sqft · 3 beds · 8 years old · 2.5 km from the metro → ₹25,134 a month. Five multiplications and one addition. That is all a prediction ever is.

THE TRAP

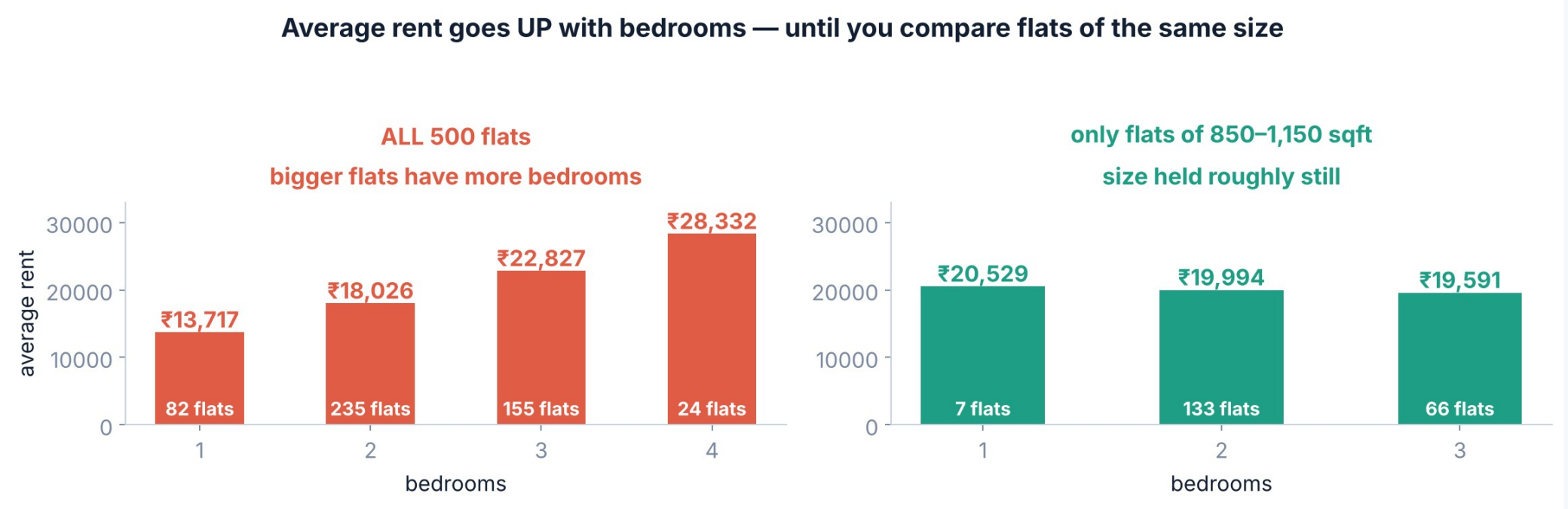
The same column, opposite signs



Fit bedrooms on its own and it says +₹4,683. Fit it alongside sqft and it says −₹1,474. Nothing was changed in the data. No bug. Both numbers are correct.

THE EXPLANATION

Because the two questions are not the same question



Ask “do 3BHK flats cost more than 2BHK?” and the answer is yes — because they are bigger. Ask “at 1,000 sqft, is a 3BHK worth more than a 2BHK?” and the answer is no — the rooms are poky.

THE RULE

What that flip actually teaches you

A COEFFICIENT IS NOT A PROPERTY OF THE COLUMN

It is a property of the column AND every other column in the model. Change the company it keeps and the number changes.

SO ALWAYS SAY WHAT ELSE IS IN THE MODEL

“Bedrooms are worth −₹1,450” is meaningless on its own. “−₹1,450, once area is accounted for” is a finding.

This has a name outside machine learning: **Simpson's paradox**. A pattern that holds for the whole group can reverse inside every subgroup. It is behind a great many wrong headlines, and multiple regression is the standard tool for not falling for it.

This is why multiple regression exists. One column at a time tells you what LOOKS true. All of them together tells you what is still true once the obvious explanation is taken out.

SECTION 3 OF 5

When columns overlap

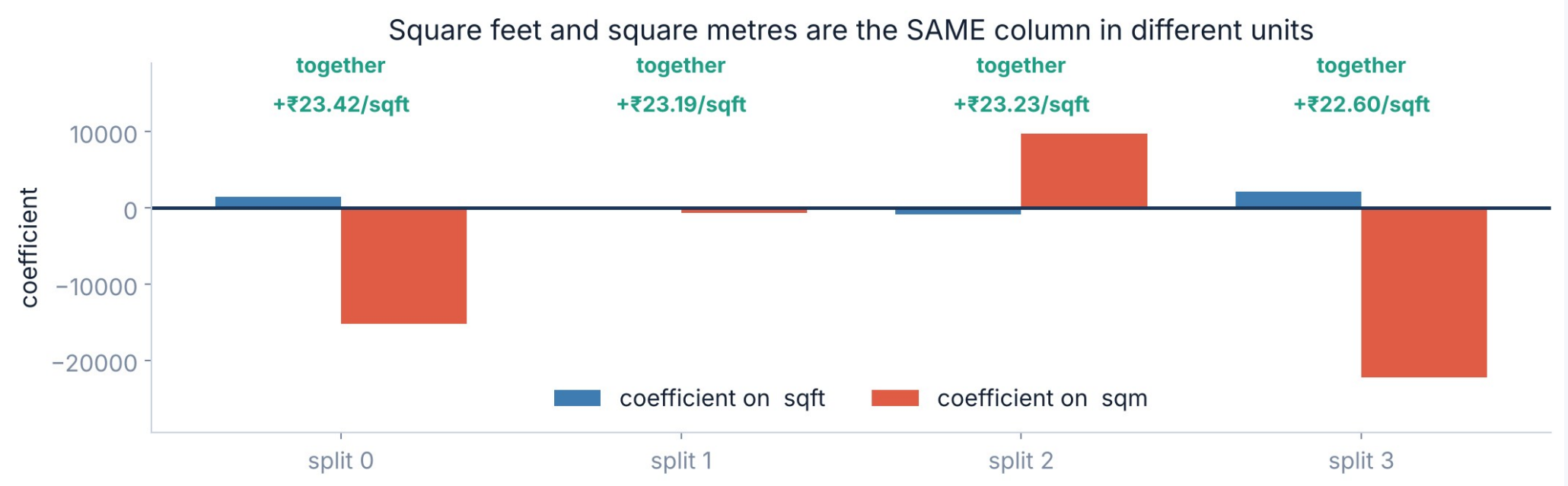
Two columns saying the same thing confuse the answer.

- The same column twice
- Coefficients go wild
- The score is fine
- What to do

THE EXTREME CASE

What if two columns say exactly the same thing?

EXAMPLE **Somebody adds a sqm column — the area again, in square metres. Nothing new, just divided by 10.76.**



Four different splits of the same data. The two coefficients swing from +1,435 to −883. But look at the top row: their COMBINED effect is +₹23 per sqft every single time.

WHAT IS ACTUALLY DAMAGED

The predictions are fine. The explanation is ruined.

	without sqm	with sqm	damaged?
R^2 on hidden rows	0.811	0.811	no
average miss	₹2,288	₹2,288	no
coefficient on sqft	+23.41	swings ±2,000	RUINED
can you explain it?	yes	no	RUINED

So does it matter? If you only want predictions — barely. If you want to tell somebody what drives rent, it matters completely, because your headline number is now whatever the split happened to pick.

The textbook name is MULTICOLLINEARITY. The plain version: two columns are telling the model the same thing, so it cannot decide which one to thank.

THE FIX

How to spot it, and what to do

1 SPOT IT

Correlation between every pair of input columns. Anything above about 0.8 is worth a look.

2 OR SPOT IT

Refit on a different split. If a coefficient swings wildly, two columns are competing.

3 DROP ONE

Keep whichever is easier to explain. sqft or sqm — never both.

4 OR COMBINE

Two exam marks that move together? Use their total instead of both.

```
flats[["sqft", "bedrooms", "age_years", "metro_km"]].corr().round(2)

#  sqft and bedrooms correlate 0.79  ->  high, but they are NOT the same thing,
#  and keeping both is exactly what revealed the bedroom effect. Judgement, not a rule.
```

0.79 between sqft and bedrooms did not stop us — dropping bedrooms would have thrown away the most interesting finding of the day. Correlation is a flag to investigate, not a rule to obey.

SECTION 4 OF 5

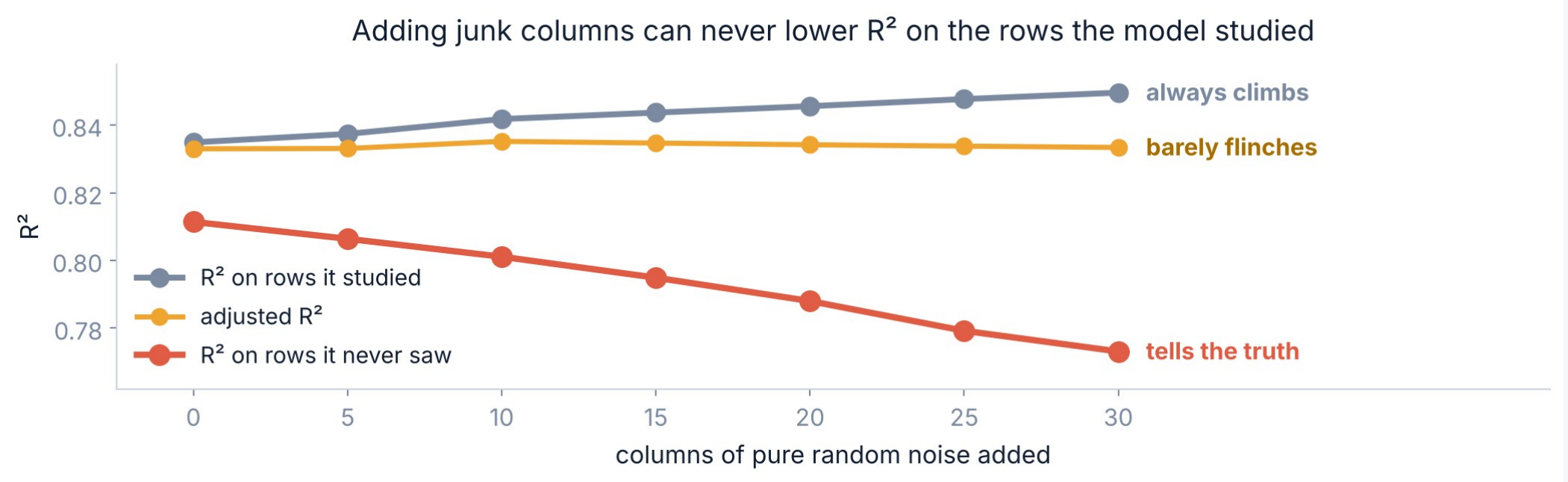
Is a column worth it?

R^2 will always say yes. R^2 is lying.

- R^2 can only go up
- Adjusted R^2
- The honest test
- One at a time

THE PROBLEM

R^2 can never go down when you add a column



Thirty columns of pure random numbers. R^2 on the studied rows rose every single time. The model is not learning — it is being handed more ways to bend around the rows it already saw.

WHICH TO TRUST

Three ways to judge an extra column

PLAIN R^2

Always goes up. Tells you nothing about whether the column helped. Never use it to compare models with different numbers of columns.

ADJUSTED R^2

Subtracts a penalty for each column. Better — but with 350 rows it barely moved: 0.8330 → 0.8334 while thirty junk columns were added.

HIDDEN-ROW R^2

Fell from 0.811 to 0.773, steadily, exactly as it should. This is the one that answers the question you actually asked.

Adjusted R^2 is a weak guard, not a safe one. It stopped the automatic climb and then barely flinched while the honest score fell by 0.04. Hide rows and measure.

THE METHOD

Add columns one at a time, and watch



All four earned their place. Note how uneven it is: bedrooms bought +0.02, but age bought +0.10. You would never have known which was which by adding them together.

SECTION 5 OF 5

Doing it for real

Word columns, fair comparisons, and the usual traps.

- One-hot, again
- Comparing coefficients
- Checking the errors
- Four ranges at once

DAY 31, AGAIN

A word column: back to one-hot encoding

EXAMPLE

We go back and collect one more fact about each flat: is it furnished, semi-furnished, or unfurnished?

furnishing · one word column becomes two yes/no columns

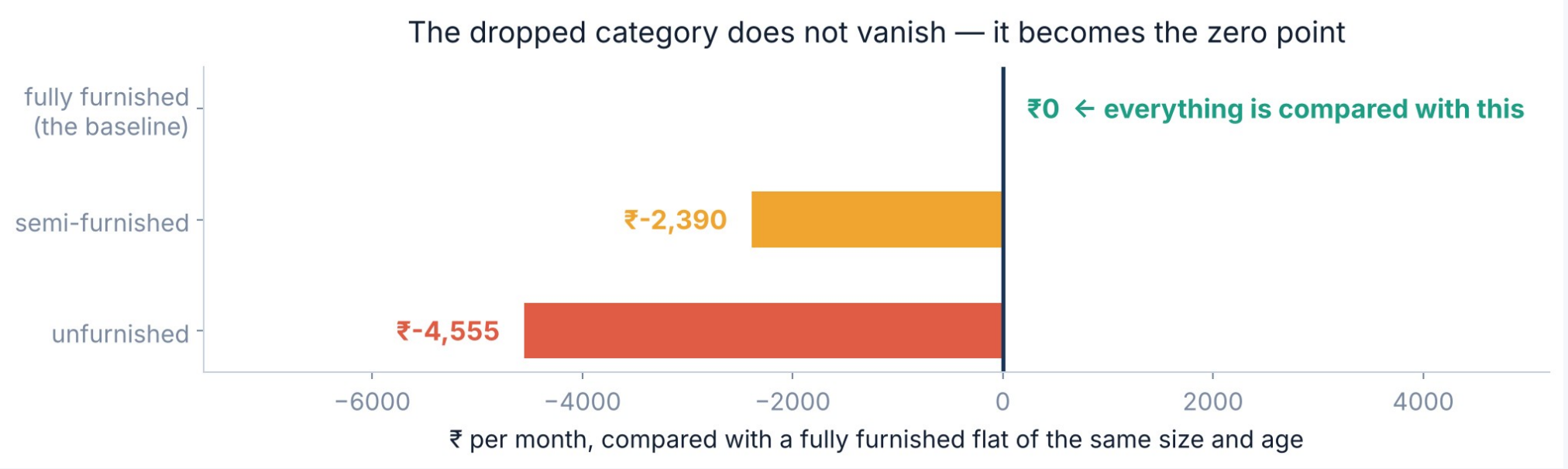
furnishing	furnishing_semi	furnishing_unfurnished
full	0	0
semi	1	0
unfurnished	0	1

Three categories, TWO columns. "full" is the one left out — it is the baseline every coefficient is measured against.

Three categories become TWO columns, not three. One category is deliberately left out, and drop_first=True in pandas is what does it.

THE BASELINE

Reading a one-hot coefficient



Every furnishing coefficient is measured against the one that was dropped. Change which category you drop and all the numbers change — but every PREDICTION stays identical.

MEASURE, DO NOT ASSUME

Did the furnishing column earn its place?

0.772
four columns, no furnishing

→

0.830
with furnishing added

→

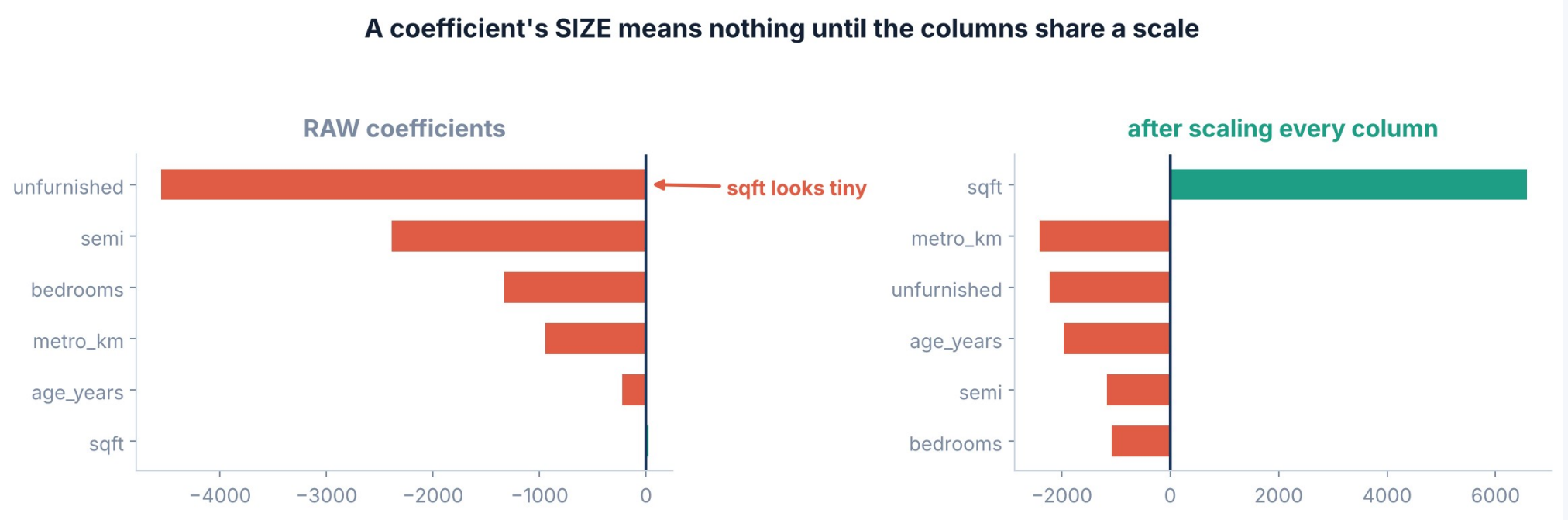
+0.058
the gain, on hidden rows

And in money: the average miss fell from ₹2,636 a month to ₹2,278 — about ₹360 better per flat. That is the number to put in front of somebody who does not know what R^2 is.

the question	the answer that convinces people
Is the model any good?	It is wrong by about ₹2,278 a month on flats it has never seen
Is it better than nothing?	Guessing the average is wrong by far more — compare, always
Was that column worth collecting?	It cut the average miss by ₹360 per flat

COMPARING COLUMNS

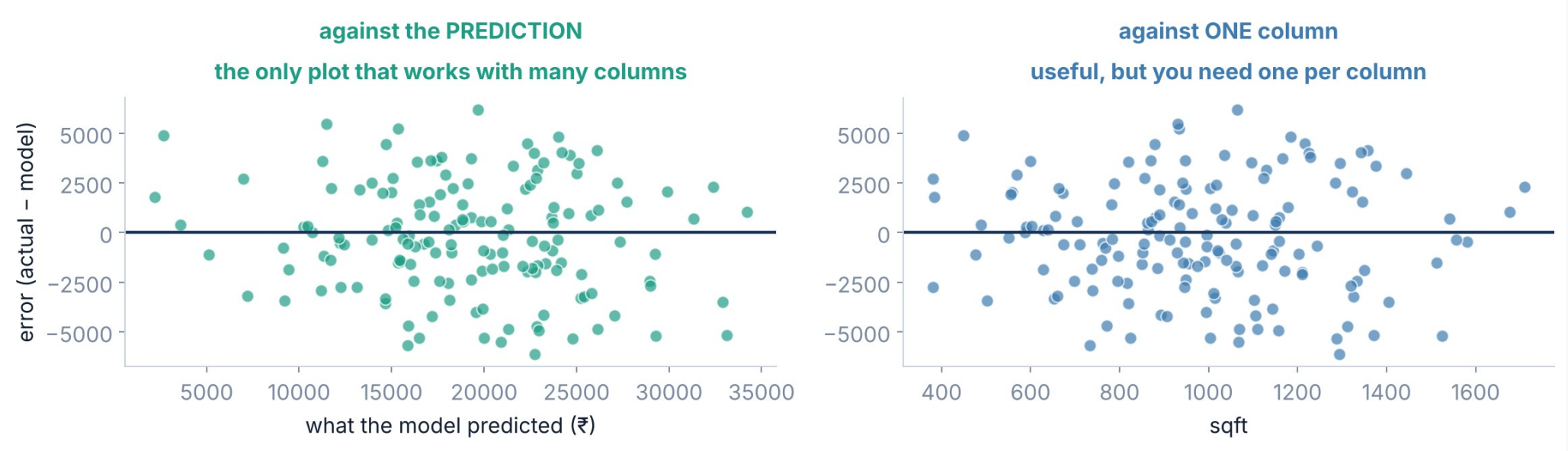
Which column matters most? Not the biggest coefficient.



Raw, sqft looks like the smallest effect in the model at +23. Scaled, it is three times bigger than anything else. A coefficient's size depends entirely on the units of its column.

THE RESIDUAL PLOT, UPDATED

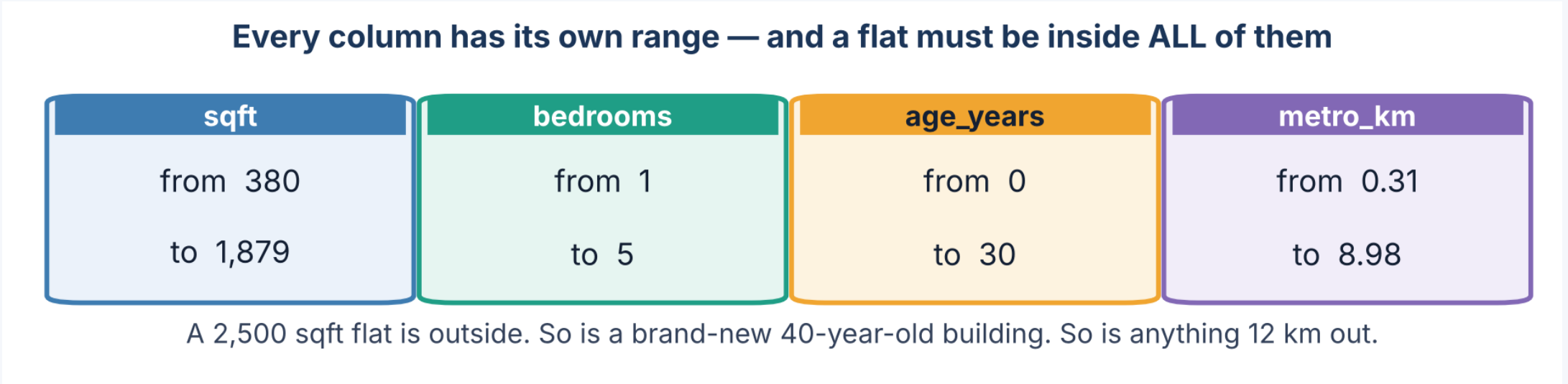
Checking the errors, with four columns



With one column you plotted errors against that column. With four, plot them against the PREDICTION — one chart that covers the whole model at once.

EXTRAPOLATION, AGAIN

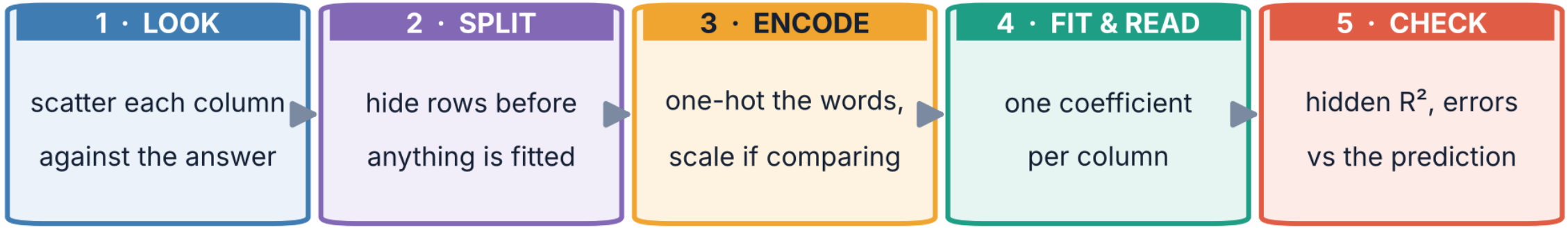
Four columns means four ways to walk off the edge



And a flat can be inside every single range and still be impossible: 400 sqft with 5 bedrooms. No row like that existed, so the model has never seen the combination.

PUT IT TOGETHER

The five steps, updated for many columns



Step 1 grew. With one column you drew one scatter plot. With six you draw six — plus the correlation table between the inputs, which is what catches two columns secretly saying the same thing.

BEFORE YOU GO

Five mistakes everybody makes once

#	the mistake	what you will see	the fix
1	Reading a coefficient on its own	“Bedrooms are worth −₹1,450”	Always say what else is in the model
2	Comparing raw coefficient sizes	“sqft barely matters, it is only 23”	Scale the columns first
3	Judging a column by plain R^2	Every column looks like it helped	Hide rows and measure there
4	Keeping two columns that mean the same	Coefficients that swing between splits	Drop one, or combine them
5	Predicting an impossible combination	A confident number for a 400 sqft 5BHK	Check the combination, not just each range

Number 1 is the one that ends up in reports. A coefficient without its context is not a finding — it is a number that happens to be true of one particular model.

SUMMARY

What to remember from today

One number per column	answer = $w_1x_1 + w_2x_2 + \ldots$ + intercept. Nothing else about yesterday changed.
Coefficients hold the rest still	"+₹23 per sqft" compares two flats identical in every other column. That clause is not optional.
A sign can legitimately flip	Bedrooms: +₹4,683 alone, −₹1,474 beside area. Both correct. They answer different questions.
Overlapping columns ruin the story	sqft and sqm left the predictions untouched and made the coefficients meaningless. Drop one.
R^2 always rises	Thirty junk columns raised it every time. Adjusted R^2 barely flinched. Only the hidden rows told the truth.
Scale before you compare	Raw, sqft looked like the smallest effect. Scaled, it was three times the next one.

19 QUESTIONS

Tonight's practice

A **Work it out**
Predict by hand, read coefficients, and explain a sign flip
8 questions

B **Find the mistake**
Four programs: a misread coefficient, a duplicate column, a fake gain
4 questions

C **Write the code**
Build, encode, scale, compare and diagnose a six-column model
5 tasks

D **Challenge**
Build it from scratch, then dismantle a model that blames the wrong thing
2 tasks

Part D hands you a hospital model that blames nurses for patient deaths, and asks you to find the column that explains it.

−₹1,474 per bedroom

…once you hold the area still. A number that is wrong on its own, correct in context, and useless unless you say which. That is multiple regression.

NEXT · DAY 35

Logistic regression — the same machinery, but the answer is yes or no.

Slides & notebooks by **Srinivasa Sai Chava** · Boston University